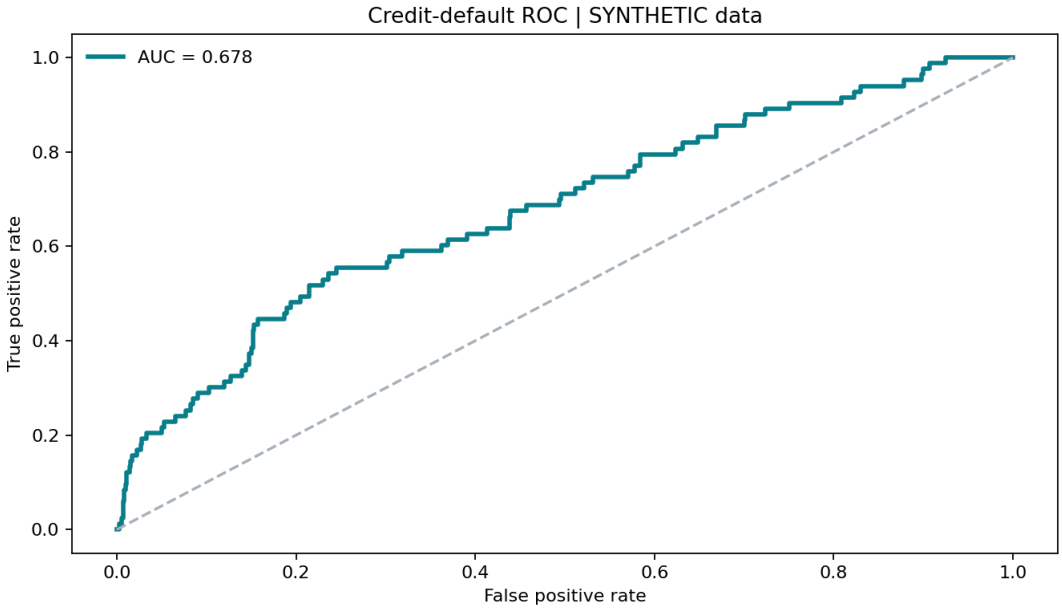
# Credit Risk | Learning Case

**DEMONSTRATION ONLY - SYNTHETIC DATA** | Logistic regression, never operational lending decisions

| | |
|---|---|
| Demo sample | **3,200 synthetic rows; 800 held-out test rows** |
| ROC-AUC on SYNTHETIC test | **0.678 \| Average precision 0.242** |
| Observed synthetic default prevalence | **10.4% \| Brier score 0.087** |
| Exploration threshold | **0.20 probability; not an approved credit policy** |




## Method

The workflow splits rows into stratified 75% train and 25% test samples. Median imputation and feature scaling are fitted on training observations only; a simple logistic regression then generates held-out default probability estimates.

AUC measures ranking discrimination, not calibration. The illustrative 0.20 cutoff trades precision and recall, and is not a proposed lending rule. Synthetic labels were created from a known probabilistic formula.

## Where real data fits

The companion `src/analyze.py` is written to accept the official UCI Default of Credit Card Clients Excel file (30,000 historical Taiwanese records). **That dataset was not available in this working environment and was not run to produce this brief.** Download the official UCI .xls and use `--mode uci` to reproduce real-data metrics.

For the real-UCI baseline, SEX, EDUCATION and MARRIAGE are excluded. Removing them does not demonstrate fairness; a meaningful credit system would require documentation, bias assessment, fresh data, out-of-time validation and governance.

## Responsible interpretation

The plotted curve and the displayed metrics are **synthetic demo metrics only**; they must not be cited as empirical findings about the UCI dataset or any Canadian borrower population.

**Source:** Optional real dataset: Yeh, I-C. (2009), UCI Default of Credit Card Clients (CC BY 4.0), https://doi.org/10.24432/C55S3H